

Two-Step Gated GRPO for Long-Horizon Trajectories

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Abstract

We specify a two-step gated GRPO method for long-horizon code-agent trajectories. At each observed decision state, terminal rewards are averaged within each distinct action before equally weighted action means are standardized. A visit with near-zero first-step advantage receives an additional signal only if its action leads to a child state that passes a dispersion gate and a second-action selection. Each decision-step advantage is assigned to every model-generated token of that step, with token-level clipping and KL regularization. This document specifies a theoretical procedure; it does not report an implementation or experiment.

1 Problem, data, and motivation

Let π_{old} be the known behavior policy, τ_i an old trajectory, and R_i its terminal reward. Let $s_{i,j}$ and $a_{i,j}$ be the complete decision state and edit or tool-call action at position j of trajectory i . A symbol such as s_t denotes a state under analysis; different trajectories may visit it at different position indices. The aim is to identify first actions whose average terminal outcomes are near the state’s action-mean average, but whose child states distinguish favorable from unfavorable next actions.

For intuition, with $h = (s_t, a)$ and $u = a'$, the law of total variance gives

$$\text{Var}(R \mid h) = \text{Var}_{u|h}(\mathbb{E}[R \mid h, u]) + \mathbb{E}_{u|h}[\text{Var}(R \mid h, u)]. \quad (1)$$

Our dispersion gate measures the spread of *equally weighted distinct action-mean rewards* at parent and child states. It does not separately gate within-action outcome variation or add a branch-reliability filter. Because the displayed variance identity weights actions by their distribution, its first term is not generally equal to the dispersion used below.

2 Action-mean advantage at a state

For any observed state s , collect decision *visits*, not unique trajectory identifiers:

$$\mathcal{I}(s) = \{(i, j) : s_{i,j} = s\}, \quad K_s = |\mathcal{I}(s)|, \quad \mathcal{U}(s) = \{a_{i,j} : (i, j) \in \mathcal{I}(s)\}. \quad (2)$$

If one trajectory revisits the same complete state, each visit is counted. Such visits can share R_i ; they are separate decisions, not independent trajectories. For $u \in \mathcal{U}(s)$, let

$$\mathcal{I}(s, u) = \{(i, j) \in \mathcal{I}(s) : a_{i,j} = u\}, \quad n_s(u) = |\mathcal{I}(s, u)|, \quad q_s(u) = \frac{1}{n_s(u)} \sum_{(i,j) \in \mathcal{I}(s,u)} R_i. \quad (3)$$

Thus repeated visits by one trajectory can repeat its terminal reward in an action mean. Different observed actions have equal weight in the subsequent normalization, irrespective of their visit counts:

$$d_s = |\mathcal{U}(s)|, \quad \bar{q}_s = \frac{1}{d_s} \sum_{u \in \mathcal{U}(s)} q_s(u), \quad \sigma_{q,s} = \left[\frac{1}{d_s} \sum_{u \in \mathcal{U}(s)} (q_s(u) - \bar{q}_s)^2 \right]^{1/2}. \quad (4)$$

The action-mean advantage is returned to every visit with that action:

$$A_{i,j}^{\text{state}} = \begin{cases} \frac{q_s(a_{i,j}) - \bar{q}_s}{\sigma_{q,s}}, & \sigma_{q,s} > 0, \\ 0, & \sigma_{q,s} = 0. \end{cases} \quad (i, j) \in \mathcal{I}(s). \quad (5)$$

The zero-variance rule applies to the advantage at this position, not to the entire trajectory. If only one distinct action appears, this rule also sets its advantage to zero. Equal weighting of distinct actions is used only for the normalization; all sampled visit positions remain in the training objective, without an extra $1/n_s(u)$ correction. This A^{state} differs from original outcome GRPO, which normalizes rewards across sampled trajectories [1].

3 Two selection rounds and the dispersion gate

At a parent state s_t , use the same action-mean standardization for the first-round advantage and dispersion. Define

$$\mathcal{I}_t(s_t) = \mathcal{I}(s_t), \quad K_t = |\mathcal{I}_t(s_t)|, \quad \sigma_t = \sigma_{q,s_t}, \quad A_{1,i,j} = A_{i,j}^{\text{state}} \quad ((i, j) \in \mathcal{I}_t(s_t)). \quad (6)$$

If $\sigma_t = 0$, the state creates no gated candidate and its fallback advantages are zero. Otherwise set

$$M_t = \lceil 0.2K_t \rceil. \quad (7)$$

Let $\mathcal{L}_t(s_t)$ contain exactly the M_t visit positions with the smallest $|A_{1,i,j}|$. Include all positions strictly below the cutoff, then fill remaining places uniformly without replacement from all positions tied at the cutoff. A trajectory revisiting s_t can have one visit selected and another unselected. Candidate actions are deduplicated only after this visit selection:

$$\mathcal{C}_t(s_t) = \{a_{i,j} : (i, j) \in \mathcal{L}_t(s_t)\}. \quad (8)$$

For each candidate a , define its selected and total visit counts by

$$m_t(a) = |\{(i, j) \in \mathcal{L}_t : a_{i,j} = a\}|, \quad n_t(a) = |\{(i, j) \in \mathcal{I}_t : a_{i,j} = a\}|. \quad (9)$$

For example, selected actions $[a, a, a, b]$ give one candidate a with $m_t(a) = 3$, not three separate child-gate computations. If the gate passes, each of those three selected positions can receive one λ increment; an unselected occurrence of the same a does not.

For each candidate a , set $s_{t+1} = (s_t, a)$. Use *all* old visits reaching this child state, including positions not selected in the first round. Let $\mathcal{I}_{t+1} = \mathcal{I}(s_{t+1})$, and for each observed next action u compute

$$\mathcal{I}_{t+1,u} = \{(i, j) \in \mathcal{I}_{t+1} : a_{i,j} = u\}, \quad q(u) = \frac{1}{|\mathcal{I}_{t+1,u}|} \sum_{(i,j) \in \mathcal{I}_{t+1,u}} R_i. \quad (10)$$

Thus $q(u) = q_{s_{t+1}}(u)$. Require at least five distinct observed next actions and nonzero child dispersion:

$$d = |\mathcal{U}(s_{t+1})| \geq 5, \quad \sigma_{t+1} = \sigma_{q,s_{t+1}} > 0. \quad (11)$$

The parent and child dispersions are both equally weighted standard deviations of distinct action-mean rewards, not standard deviations of individual trajectory returns. The dispersion gate is

$$G_\Delta(s_t, a) = \mathbf{1}\{\sigma_{t+1} \geq \omega\sigma_t\}, \quad (12)$$

where ω is an unspecified tunable constant. When the gate passes, compute the signed second-step advantage

$$A_2(s_{t+1}, u) = \frac{q_{s_{t+1}}(u) - \bar{q}_{s_{t+1}}}{\sigma_{t+1}}. \quad (13)$$

Let $\mathcal{S}(s_t, a)$ contain exactly the $m(d) = \lceil d/2 \rceil$ distinct actions with largest $|A_2(s_{t+1}, u)|$. Fill cutoff ties uniformly without replacement. Selected actions can have either sign, and no extra positive/negative quota or minimum absolute-advantage threshold is imposed. If any preceding check fails, define $\mathcal{S}(s_t, a) = \emptyset$.

4 Gates and two-step advantages

The action-level gate and visit-level first-step gate are

$$G_t^{\text{act}}(s_t, a) = \mathbf{1}\{\sigma_t > 0\} \mathbf{1}\{a \in \mathcal{C}_t(s_t)\} \mathbf{1}\{\mathcal{S}(s_t, a) \neq \emptyset\}, \quad (14)$$

$$B_{i,t} = \mathbf{1}\{(i, t) \in \mathcal{L}_t(s_t)\} G_t^{\text{act}}(s_t, a_{i,t}). \quad (15)$$

Thus $B_{i,t} = 1$ implies that at least one successor action passed the second round. A same- a visit outside $\mathcal{L}_t$ does not receive the first-step increment

$$\Delta A_{i,t}^{(1)} = \lambda B_{i,t}, \quad \lambda > 0. \quad (16)$$

Repeated visits by one trajectory can have different values of $B_{i,t}$. If the gate passes, each selected visit receives one increment, whereas an unselected visit taking the same action retains its A^{state} fallback unless it has a separate overlapping role.

For a passing child state and a non-overlapping second-step position, define $I_{i,t+1}^{\mathcal{S}} = \mathbf{1}\{a_{i,t+1} \in \mathcal{S}(s_t, a_{i,t})\}$. The second-step coefficient is

$$\tilde{A}_{i,t+1} = [1 + (\beta - 1)I_{i,t+1}^{\mathcal{S}}] A_2(s_{t+1}, a_{i,t+1}), \quad \beta > 1. \quad (17)$$

Since $A_{i,t+1}^{\text{state}} = A_2(s_{t+1}, a_{i,t+1})$ at this child state, unselected actions retain their ordinary action-mean advantage. Both quantities use all old visits at the child state and the same action partition. Selected positive and negative actions both receive a larger absolute coefficient. The first-step λ is not broadcast to the child action.

If one position is the previous chain’s second action and the next chain’s selected first action, the confirmed overlap rule adds their two roles:

$$\hat{A}_{i,j}^{\text{overlap}} = A_{i,j}^{\text{state}} + \lambda + \beta I_{i,j}^{\mathcal{S}} A_2(s_{i,j}, a_{i,j}). \quad (18)$$

When $I_{i,j}^{\mathcal{S}} = 1$, the action-level part is $(1 + \beta)A_2$, because the two roles are added. When $I_{i,j}^{\mathcal{S}} = 0$, it is A_2 , and the position still receives λ through its own selected first-step role.

5 Per-position advantage and GRPO update

At each trajectory position, let

$$P_{i,j} = G_{j-1}^{\text{act}}(s_{i,j-1}, a_{i,j-1}), \quad I_{i,j}^{\mathcal{S}} = \mathbf{1}\{a_{i,j} \in \mathcal{S}(s_{i,j-1}, a_{i,j-1})\}. \quad (19)$$

Take a nonexistent previous gate at a trajectory boundary to be zero. Abbreviate $A_{2,i,j} = A_2(s_{i,j}, a_{i,j})$ when $P_{i,j} = 1$. The complete training advantage is

$$\hat{A}_{i,j} = \begin{cases} A_{i,j}^{\text{state}}, & P_{i,j} = 0, B_{i,j} = 0, \\ A_{i,j}^{\text{state}} + \lambda, & P_{i,j} = 0, B_{i,j} = 1, \\ \beta A_{2,i,j}, & P_{i,j} = 1, B_{i,j} = 0, I_{i,j}^{\mathcal{S}} = 1, \\ A_{i,j}^{\text{state}}, & P_{i,j} = 1, B_{i,j} = 0, I_{i,j}^{\mathcal{S}} = 0, \\ \hat{A}_{i,j}^{\text{overlap}}, & P_{i,j} = 1, B_{i,j} = 1. \end{cases} \quad (20)$$

One edit or tool call is one decision step j . Its training advantage is computed from full-trajectory terminal rewards, not an immediate environment reward. If step j contains $L_{i,j}$ model-generated tokens, every such token—including thinking text and edit or tool-call content—receives the same advantage:

$$\hat{A}_{i,j,k} = \hat{A}_{i,j}, \quad k = 1, \dots, L_{i,j}. \quad (21)$$

Prompts, environment observations, and tool responses are not policy prediction targets. They can still enter the conditioning context. Let $y_{i,j,k}$ be generated token k , and let $h_{i,j,k}$ be its full preceding context. For the known behavior policy π_{old} , compute and clip each token ratio separately:

$$r_{i,j,k}(\theta) = \frac{\pi_{\theta}(y_{i,j,k} \mid h_{i,j,k})}{\pi_{\text{old}}(y_{i,j,k} \mid h_{i,j,k})}, \quad (22)$$

$$\ell_{\text{clip}}(r, A) = \min\{rA, \text{clip}(r, 1 - \epsilon, 1 + \epsilon)A\}. \quad (23)$$

The KL penalty is enabled and also computed for each generated token, using the sampled-token form of GRPO [1]:

$$D_{i,j,k}^{\text{KL}}(\theta) = \frac{\pi_{\text{ref}}(y_{i,j,k} \mid h_{i,j,k})}{\pi_{\theta}(y_{i,j,k} \mid h_{i,j,k})} - \log \frac{\pi_{\text{ref}}(y_{i,j,k} \mid h_{i,j,k})}{\pi_{\theta}(y_{i,j,k} \mid h_{i,j,k})} - 1. \quad (24)$$

First average both terms over the generated tokens in each step, then average over steps within each trajectory and finally over trajectories. With $\kappa > 0$, maximize

$$J(\theta) = \mathbb{E}_{\{\tau_i\} \sim \pi_{\text{old}}} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{j=1}^{T_i} \frac{1}{L_{i,j}} \sum_{k=1}^{L_{i,j}} \left(\ell_{\text{clip}}(r_{i,j,k}(\theta), \hat{A}_{i,j}) - \kappa D_{i,j,k}^{\text{KL}}(\theta) \right) \right]. \quad (25)$$

Equivalently, minimize $-J(\theta)$. Old rewards, gates, and advantages are fixed during an update; gradients pass through the current-policy token ratios and KL terms. Every old visit retains its own step term, including repeated visits by one trajectory. Each step has weight $1/(NT_i)$, and each generated token in that step has weight $1/L_{i,j}$; no inverse action-frequency correction is made. Thus a longer generated step does not receive a larger total weight merely for containing more tokens. The clip width ϵ , KL weight κ , and second-step amplifier β are distinct. The numerical ϵ and κ , and the choice of π_{ref} , remain unspecified.

6 Calculation order

Collect old trajectories and terminal rewards from the known π_{old} . Group all visits, including repeated visits within one trajectory, by complete state and next action. Average terminal rewards within each action and equally standardize the distinct action means to obtain A^{state} . If this dispersion is zero, use fallback advantage zero and do not create first-round candidates for that parent; token KL still applies.

At each eligible parent select exactly $M_t = \lceil 0.2K_t \rceil$ visits by smallest $|A_1|$, resolving cutoff ties by uniform sampling without replacement. Deduplicate their actions only after selection. For each candidate action, use all visits at its child state and require at least five distinct next actions, nonzero child dispersion, and $\sigma_{t+1} \geq \omega\sigma_t$. Select exactly $\lceil d/2 \rceil$ distinct next actions by largest $|A_2|$, again sampling cutoff ties. Assign the per-position advantage by the five cases above, broadcast it to that step’s model-generated tokens, and update θ using tokenwise clipping and KL with step/trajectory averaging.

7 Open experimental choices and theoretical limitations

This is a theoretical procedure, not a specified experiment. The numerical values and search ranges for $\omega, \lambda, \beta, \epsilon, \kappa$, the reference policy, data split, sampling budget, failure-sample handling, stopping conditions, and evaluation metrics remain open. No training run depending on them is implied.

The first-round near-zero criterion is based on action-mean advantages and visit counts. Cutoff ties can select only some visits with the same action; repeated visits by one trajectory can be treated differently. With two distinct actions and unequal means, their equally standardized absolute advantages tie. With zero action-mean dispersion, no first-round candidate is formed even if rewards vary within each action. The ceiling in M_t can produce a selection fraction well above 20% at small K_t ; no minimum K_t is imposed.

All visits with the same state and action share A^{state} , erasing their different terminal outcomes at that position; later positions may receive different advantages at their respective states. Child dispersion reflects differences between observed next-action means, not within-action reward mixtures. A rare branch mean may be noisy, but no reliability gate is specified. Parent and child may have different numbers of distinct actions, so the stability of comparing their equally weighted dispersions should be evaluated experimentally. Selection by $|A_2|$ may omit one sign or include a zero-advantage action; no sign quota or positive minimum threshold is imposed. Raising the probability of the first action with λ can raise the probability of a harmful

full chain even if a negative-advantage successor becomes conditionally less likely; this requires separate empirical checking.

References

- [1] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. arXiv:2402.03300.